

Global Considerations in Kidney Disease: The Pacific Islands and Islands of the Caribbean

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CHAPTER OUTLINE

OCEANIA, 2311

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KEY POINTS

- The burden of chronic kidney disease is increasing in the small island developing states, largely driven by growth in noncommunicable chronic illnesses such as diabetes mellitus.
- The burden of chronic kidney disease in the Caribbean is rapidly increasing and is currently the eighth leading cause of death.
- Chronic kidney disease is now the fourth leading cause of death across Oceania.
- For indigenous populations, there is a need to recognize the impact of colonization and racism and prevent deficit framing to improve outcomes for chronic kidney disease.
- Access to health care, ranging from primary preventative management to kidney replacement therapy, is limited in many nations of the Oceania region outside of Australia and New Zealand, especially for those who are unable to afford care.
- Within the Caribbean, variability of health care access across countries exists, with those with larger populations and a higher social developmental index (e.g., Cuba, Dominican Republic and Puerto Rico) having more robust health care infrastructure and personnel, compared with other island countries, such as Haiti, that have limited health care infrastructure.
- In response to growth in chronic kidney disease prevalence, countries such as Fiji and Samoa are currently in the process of expanding their capacity to care for these patients with financial support from their government.
- The impact and burden of CKD in the Caribbean are likely underestimated due to limited population-based data on prevalence and outcomes in CKD, in part due to lack of surveillance systems, access to electronic records, and registry data in many Caribbean countries.

Oceania and the Caribbean host an ethnically diverse population. Several socioeconomic, environmental, and demographic factors adversely affect kidney health across the islands in the Pacific and the Caribbean, more aptly termed the Small Island Developing States (SIDS). The SIDS comprise 39 states and 18 associate members of United Nations regional commissions,⁴ which span the global extremes of resources and development from Singapore to Haiti.⁵ Given their size, remoteness, small economies, and susceptibility to environmental shocks, specific approaches are required to optimize health and delivery of health

care.⁶ In congruence with increases in noncommunicable diseases, the burden and impact of chronic kidney disease across the SIDS are high. Despite this, there are limited data on population-based chronic kidney disease (CKD) prevalence estimates and outcomes in many of these countries. The capacity for evaluation and treatment of kidney disease remains heterogeneous, with countries with more robust health care systems having improved access. What is known and unknown about kidney health and access to care across the SIDS is described here, with an in-depth discussion focusing on SIDS in Oceania and the Caribbean

where data are available. Aspects of kidney health and care will also be described across Australia and New Zealand (NZ) to highlight within-country inequities as a legacy of colonialism, which echoes across most SIDS.

OCEANIA

Oceania is a collective name for the land masses scattered throughout most of the Pacific Ocean. It is the term originally conceived by the French explorer Dumont d'Urville (1790–1842) in 1812 as “Océanie,” which is a French word derived from the Latin word *oceanus*, meaning ocean. Oceania, in its most commonly defined terms, includes >10,000 islands, with a total area of 8.526 million km² and estimated population of >40 million people. Oceania can be divided into four geopolitical subregions: Australasia, Melanesia, Micronesia, and Polynesia (Fig. 79.1), formed by 14 countries and 25 sovereign states and dependent territories including Australia (population, 25.7 million), Papua New Guinea (9.4 million), NZ (5.1 million), and other island nations with populations ranging from <100 to >450,000.⁷

Indigenous Australians are the original inhabitants of the Australian continent and nearby islands, whereas the various indigenous people of Oceania arose from migration across the Pacific. Over the years, many cultures have been influenced by migration, invasion, and colonization with

resultant diversity in cultures, economy, language, and religions. Increased influence of the Western world in the region of Oceania has led to an increased urbanization and witnessed a growing burden of adverse health outcomes including obesity, hypertension, diabetes, and CKD, as well as inequalities in accessing health care.

Kidney disease is one of the major socially transmitted diseases that contributes to higher health care utilization, poor quality of life, and serious adverse outcomes including cardiovascular disease (CVD), need for kidney replacement therapy (KRT), and death.^{8,9} Although the exact incidence and prevalence across the Oceania region are incompletely known, reported deaths and years lost due to disability resulting from CKD range up to 4.28% and 2.24%, respectively.⁷ Quantification of the burden of kidney disease is least well understood in the lowest-income countries, and given the need for higher health care utilization, there is an associated cost of kidney disease, which needs to be understood in the context of local government health care expenditure.

The per-capita income (gross domestic product [GDP] nominal per capita), not accounting for the cost of living, ranges widely in Oceania from those countries among the highest incomes in the world (Australia, US\$50,997; New Zealand, US\$44,880) to less than US\$10,000 in many smaller island countries, states, and territories. The proportion of government expenditure spent on health also ranges



Fig. 79.1 Map of Oceania region. (From Geographic Guide. <<http://www.geographicguide.com/oceania-map.htm>>. Accessed 24.01.19.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

from 4% (Papua New Guinea, Vanuatu) to 29% (French Polynesia [France Protectorate]). Unsurprisingly, there is considerable variation in health outcomes and risk factors for long-term conditions.⁷ This section discusses the burden of kidney disease, both acute and chronic, in Australia, New Zealand, and the island nations of Oceania. Given the high-income status of Australia and New Zealand, the focus in the next two sections (Australia and Aotearoa New Zealand) is more on the impact of kidney disease on the indigenous populations of both countries.

AUSTRALIA

BURDEN OF DISEASE

In Australia, >1.7 million adults older than the age of 18 years have indicators of CKD, such as reduced estimated glomerular filtration rate (eGFR) or proteinuria.¹⁰ This equates to 1 in 10 Australian adults, with even higher incidence among indigenous Australians (Aboriginal and Torres Strait Islanders) of 1 in 5¹¹ who make up approximately 3.8% of the total Australian population (estimated to be 983,700 people as of June 2021).¹² In contrast to the nonindigenous population where the CKD burden appears to be predominantly situated in major cities, the highest CKD burden in the indigenous population is found in remote and very remote areas (remote 34% vs. nonremote 13%).^{11,12} Along with remoteness, major factors contributing to these higher rates include socioeconomic deprivation, poor access to health care, and the long-term impacts of colonization and racism. With the growth in CKD burden, the number of patients with end-stage kidney failure (ESKF) has doubled in Australia over the past decade and is currently >28,000 patients.^{13,14} This growth is not surprising and is an inevitable consequence of background increase in both behavioral (e.g., inactive/insufficiently active: 56%; smoking on daily basis: 16%) and biomedical (e.g., overweight or obese: 63%; high blood pressure: 32% [including 22% with uncontrolled high blood pressure]) risk factors for CKD in the Australian general population.¹⁰ Diabetes mellitus is the fastest growing chronic disease, and diabetic nephropathy has become the most common cause of ESKF in Australia (37%) as a result.^{13,15} Inherently, the burden of CKD relates to not only the health care requirements of patients who progress to ESKF but also broader public health through its close association with CVD, which affects 22% of Australian adults.¹³ CKD was estimated to cause 1.1% of the total burden of disease and injury in Australia in 2022 with nearly 73% of this burden attributed to years of life lost due to premature death.¹³ The rate of burden was 7.8 times higher in indigenous Australians compared with nonindigenous Australians. Patients with CKD incur 85% higher health care costs and 50% higher government subsidies than individuals without CKD. In 2021 Kidney Health Australia estimated the annual cost of CKD to be A\$9.9 billion, which includes expenditure on health services, the impact on productivity, and individual costs.^{16,17} The overall cost per person living with CKD in 2021 was A\$41,795 rising to >A\$182,000 for those with kidney failure.¹⁵

DIALYSIS

According to the Australia and NZ Dialysis and Transplant (ANZDATA) Registry, there were 3281 new KRT patients in Australia in 2021 (Fig. 79.2), with an overall incidence

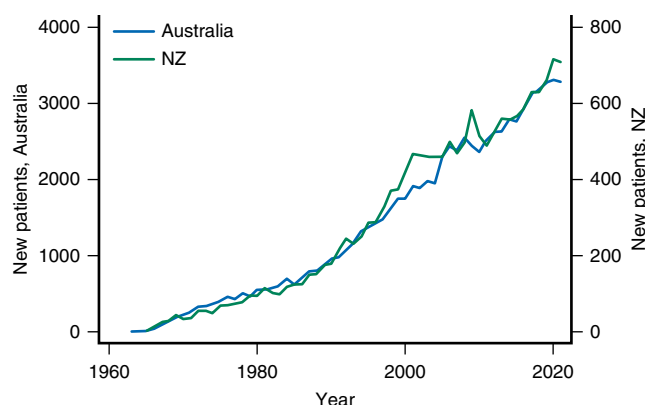


Fig. 79.2 Number of new patients commencing kidney replacement therapy from the 1960s to 2020 in Australia and New Zealand. (From ANZDATA Registry. 45th report. Chapter 1: Incidence of kidney failure with replacement therapy. Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia. <<http://www.anzdata.org.au>>; 2022.)

rate of 127 per million population (pmp).¹³ Although this rate has remained stable for several years, the number of prevalent patients has continued to climb with 28,542 (1109 pmp) patients receiving KRT at the end of 2021.¹⁴

The mean ages of female and male patients commencing KRT were 58.7 years and 61.2 years, respectively. The most common cause of ESKF among incident KRT patients in Australia is diabetes mellitus (37%), followed by glomerulonephritis (17%), hypertension (12%), polycystic kidney disease (6%), reflux nephropathy (2%), and others.¹³ In 2021, more than half (53%) of ESKF patients received dialysis, which comprised facility (hospital or satellite) hemodialysis (75%), peritoneal dialysis (PD; 17%), and home hemodialysis (8%).¹⁴ Patient survival on hemodialysis has remained relatively unchanged over the past few years.¹⁸ The ongoing costs of caring for a patient with ESKF on dialysis are significant and greatest for in-center hemodialysis (A\$76,881/person per year), followed by satellite hemodialysis (A\$63,505), PD (A\$51,640), and home hemodialysis (A\$47,775).¹⁷

Aboriginal and Torres Strait Islander Australians experience higher rates of kidney failure at all ages older than the age of 15 years (Fig. 79.3),¹⁹ with greater disparities among indigenous women.¹⁹ There are marked differences in treatment modalities for Aboriginal/Torres Strait Islanders with the majority treated with facility-based hemodialysis (74%) with few accessing home dialysis (either hemodialysis or PD) (Fig. 79.4).¹⁹ Unfortunately, this has significant impacts on the social and cultural values important to Aborigines and Torres Strait Islanders because these individuals, by dialyzing in facility units often at great distances from their homes, are removed from their people and country with major negative impacts on their mental health and well-being.

With the growth in the prevalent ESKF population, facility hemodialysis capacity has been saturated and attention has been drawn to enhancing the capacity of home dialysis, particularly PD, which makes up 70% of all home dialysis patients in Australia. Although patient survival on PD is comparable with hemodialysis (1 year 94% [95% confidence interval {CI}: 93%–95%], 3 years 75%

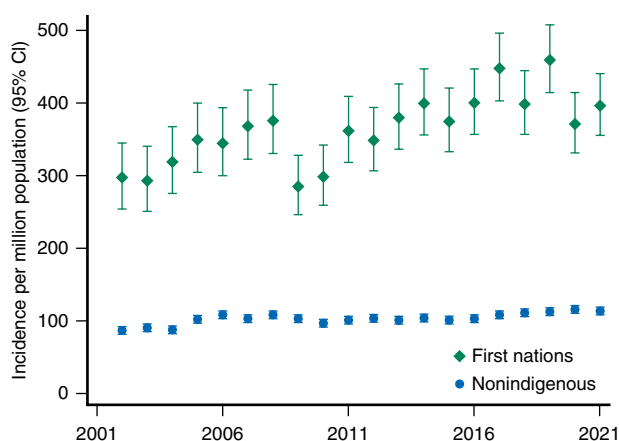
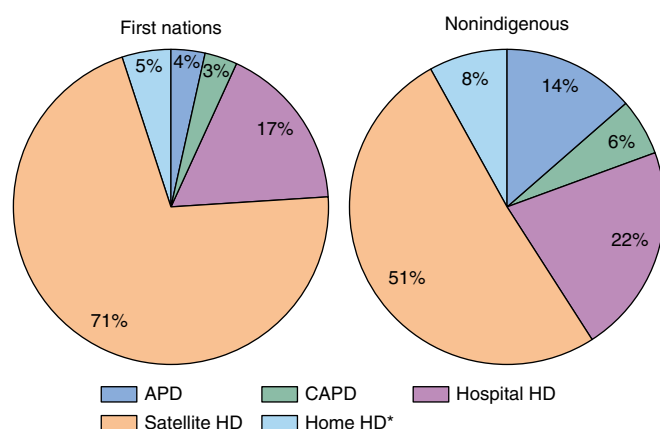


Fig. 79.3 Dialysis Modality Australia 2021. (From ANZDATA Registry. 45th report. [Chapter 2](#): Prevalence of kidney failure with replacement therapy. Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia. <<http://www.anzdata.org.au>>; 2022.)



*Includes community house HD

Fig. 79.4 Patient survival on hemodialysis at kidney replacement therapy start. Australia 2010–2020, censored for transplant and transfer to PD. (From ANZDATA Registry. 45th report. [Chapter 4](#): Haemodialysis. Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia. <<http://www.anzdata.org.au>>; 2022.)

[95% CI: 73%–76%], 5 years 55% (95% CI: 52%–56%)], this is significantly modified by the age of the patients with 5 year survival rates of 85% (95% CI: 81–88) for those <40 years compared to 50% (95% CI: 48–52) for those aged 65 to 74 years. Further uptake of PD has been hampered by relatively limited time on PD (3 years 37% [95% CI: 35%–39%]).²⁰ Time on PD (previously referred to as technique survival) has improved slowly over the years but remains relatively low and is driven by infection (19%; e.g., peritonitis) inadequate dialysis (18%; e.g., inadequate fluid ultrafiltration and solute clearance) and mechanical issues (12%). Despite improvement in existing PD guidelines, along with the development of key performance indicators to meet evidence-based practice, and reinforcement of PD training, which has led to a significant improvement in PD peritonitis rates,²⁶ there remains a large (sevenfold) variation in peritonitis rates among PD units and up to twofold variation between states. Although some of these differences relate to patient-level risk factors (e.g., living

distantly from PD unit²² and indigenous²³), the majority of the risk has been shown to be associated with modifiable center-level characteristics (e.g., center size, proportion of PD, and peritoneal equilibration test use at PD start).^{21,24}

KIDNEY TRANSPLANTATION

In 2021, 857 (33 pmp) transplants were performed in Australia, 202 of which were live-donor kidney transplants. The reduced numbers reflect the impact of the COVID pandemic as pre-COVID transplant numbers were as high as 1149 per annum (2018), which was the highest number ever performed (40 pmp, which is still only about 30% of the rate of incident ESKF patients [127 pmp]).^{12,25}

The number of renal transplants has consistently increased (2004: 650 patients; 2018: 1149 patients), largely from an increase in the rate of deceased-donor renal transplantation in the context of a national reform program “Donate Life” led by the Australian Organ and Tissue Donation and Transplant Authority (AOTDTA) to increase organ donation rate (www.donatelife.gov.au). With the growing burden of CKD and patients with ESKF requiring KRT, there has been a focus on increasing transplant rates as it confers benefits at both patient (e.g., improved survival)²⁵ and government levels (e.g., cost-effective treatment, estimated cost in the first year A\$65,375 [deceased-donor renal transplant] to A\$70,553 [live-donor renal transplant], with ongoing cost from the years thereafter of A\$10,749/year).¹⁷ An increased rate of kidney transplantation has translated into a growth in the prevalence of patients with functioning grafts to 13,349 patients (519 pmp) in 2021 compared with 6981 patients (341 pmp) in 2005.³⁰ Unfortunately, Aboriginal/Torres Strait Islanders are markedly underrepresented, making up only 3% of the prevalent transplant recipients in 2021.²⁵ Similar within-country disparities in transplant rates involving ethnic minorities have been reported previously in Canada²⁶ and the United States.²⁷ There have been specific initiatives commenced to try to address these inequalities of access with the setting up of the National Indigenous Kidney Transplantation Taskforce: changing systems to achieve equitable access to kidney transplantation. A number of culturally specific recommendations, along with developing an Aboriginal and Torres Strait Islander kidney health workforce and building effective change-enabling structures to improve the delivery of culturally safe care, have been published.^{28,29}

For first-time recipients of kidney transplants in Australia, the 1-year graft survival rate was excellent, especially for living-donor kidney transplantation recipients (99% vs. 96% in deceased-donor recipients).³⁰ The 5-year survival rates for primary transplants during 2015–2021 are 92% and 83%, respectively.³⁰ Although the long-term outcomes (i.e., >10-year graft survival) have also improved gradually, the trend has been less evident compared with short-term outcomes.^{25,30}

According to the ANZDATA Registry, the most common cause of graft loss during the years 2012–2021 was death with a functioning graft ($n = 2447$, 47.7%), followed by chronic allograft nephropathy ($n = 1320$, 25.7%). The most common cause of death was cancer ($n = 353$, 26%), followed by CVD ($n = 308$, accounting for 23% of deaths).²⁵ Relative cancer mortality is higher than the general population (standardized mortality rate [SMR] 2.9, 95% CI: 2.7–3.1) with nonmelanoma skin cancer (SMR 50.9, 95% CI:

43.5–59.6) and lymphoma (SMR 42.2, 95% CI: 35.3–50.5) the highest in transplant recipients.³¹ Although the patient survival rates are superior after transplant compared with remaining on dialysis, the risk of death remains significantly greater than in the general population by up to 10-fold.²⁵

NONDIALYSIS SUPPORTIVE CARE FOR END-STAGE KIDNEY FAILURE

The exact number and characteristics of patients who choose not to dialyze when they reach ESKF is unknown. The Patient Information about Options for Treatment (PIVOT) study observed that 14% of incident stage V CKD patients chose a no-dialysis pathway,³² but this is likely to be an underestimate of true disease burden because it did not account for patients who were never referred to nephrology services in the first instance. It has been suggested that for every patient who dies while receiving KRT, there is another patient who has died without accessing KRT.^{33,34}

ACUTE KIDNEY INJURY

Acute kidney injury (AKI) has a broad range of causes. It is commonly diagnosed in general hospitalized patients in developed countries such as Australia (e.g., critically ill patients admitted to intensive care units in the context of sepsis). Another cause of AKI relatively unique to Australia includes snakebites caused by hemotoxic or myotoxic snakes, such as brown snakes, which account for more than half of all hospitalized snakebites in Australia.^{35,37} There have been reports of thrombotic microangiopathy-like presentations after brown snakebites with resultant severe AKI requiring temporary treatment with hemodialysis and plasma exchange.³⁵

In Australia, hospitalizations due to AKI have more than doubled between 2000–2001 and 2012–2013 (from 8050 to 18,010), representing an increase of 6%/year.³⁸ More recent studies using a laboratory-based algorithm detected AKI in 12.4% of all hospitalizations in 16.4% of patients admitted to 4 metropolitan hospitals in Sydney Australia, over a 5-year period, whereas hospital coding for AKI significantly underreported AKI incidence.³⁹ Even though the deaths due to AKI have remained relatively similar over the past decade (an average of 4670 deaths/year between 2000 and 2012), the average length of stay in AKI hospitalizations was about twice as long as the average length of stay for hospitalizations overall (11.4 days vs. 5.6 days).³⁸ People living in remote areas, in socioeconomically disadvantaged areas, and with Aboriginal and Torres Strait Islander status were associated with higher rates of hospitalizations and mortality due to AKI.^{38,40} There was a high rate of infections as the primary cause of AKI with the age groups skewed to a much younger age group than the national average. Again, there was significant underreporting of AKI on the discharge summaries.³⁹

INDIGENOUS AUSTRALIANS

Disparity in the risk of renal disease and poorer outcomes are uniformly seen across all phases of disease and treatment in indigenous Australians (Aboriginal and Torres Strait Islanders).^{12,19} Incidence rates of kidney failure with replacement therapy for indigenous Australians are markedly higher (see Fig. 79.4) compared with the nonindigenous population.¹⁹ This variation is due to

number of factors including the impact of institutional racism and colonization, reduced access to treatment, as well as the impact of underlying conditions such as diabetes mellitus, rates of disease progression, and referral patterns.^{2,29} Indigenous Australians also present with ESKF at much younger ages with a peak in the 35- to 44-year-old age group.^{12,19} The majority of Aboriginal and Torres Strait Islanders commenced KRT with hemodialysis (88%) with hemodialysis incidence approximately 5-fold higher (351 pmp) compared with nonindigenous Australians (78 pmp),¹⁹ which reflects a higher proportion of patients receiving facility hemodialysis (12.7% vs. 51.0%).¹⁹ By contrast, the proportion of prevalent indigenous Australian patients with functioning transplants was only about 3% of all Australian patients (241 patients vs. 10,310 patients).^{19,25} Within indigenous Australians, the risk of CKD-related hospitalizations not involving dialysis was significantly higher in those living in remote and very remote areas (hospitalization rate ratio for indigenous vs. nonindigenous Australians: major city: 2.9; inner regional: 3.8; outer regional: 6.9; remote/very remote: 12.8).^{41,42} Previous studies have shown that remoteness of residence is associated with pervasive socioeconomic disadvantage, which includes the impact of colonialism and institutional racism, as well as including poverty, poor nutrition and food insecurity, poor housing, unemployment, and impaired access to services of all kinds.^{2,29,41,42} Furthermore, indigenous Australians were almost four times as likely to die with CKD as a cause of death than nonindigenous Australians.^{19,25}

AKI is also significantly more common in indigenous Australians, with hospitalization rates almost three times higher than those of nonindigenous Australians (males 158 vs. 71 per 100,000 population, respectively; females 229 vs. 70 per 100,000 population).⁴⁸ Similarly, AKI death rates are 1.8 times higher in indigenous Australians (35 vs. 19 per 100,000 population) and are considerably higher in indigenous females than indigenous males (2.1-fold vs. 1.6-fold higher than nonindigenous males and females, respectively).³⁸ As with CKD, there is a complex interaction among indigenous status, remoteness, socioeconomic status, as well as cultural inequities and racism that impact AKI occurrence and outcome in Australia.

ETHNIC INEQUITIES

The burden of kidney disease (AKI, CKD, and ESKF) in indigenous Australians is disproportionately greater compared with that in nonindigenous Australians (see Fig. 79.3)^{19,25} and affects these patients at an earlier age. For instance, of all the patients who commenced KRT for treatment of ESKF in Australia between the years 2010 and 2021, 59% of indigenous patients were younger than 55 years old, compared with 31% of nonindigenous patients.¹⁹ Although the most common cause of ESKF in Australia is diabetic nephropathy for both groups, the proportion of indigenous patients with diabetic nephropathy is more than double that of nonindigenous patients (69% vs. 35%).¹⁹ Differences in the causes of CKD and AKI across these two groups are more challenging to describe in detail because data are not as readily available due to a lack of established CKD and AKI registries (the ANZDATA only includes ESKF patients¹⁹ who received dialysis or kidney transplantation). The reasons for ethnic inequality in the burden of kidney

disease among indigenous Australians have been attributed to higher levels of socioeconomic disadvantage^{43,44,45} residence in “remote” and “very remote” areas of Australia¹⁹ and a high burden of chronic diseases, such as diabetes mellitus, which is an important cause of kidney disease.^{44,45} Many other contributory factors include poorer nutrition, higher tobacco use, alcohol consumption, and lower levels of education attainment, employment, social support, and housing, as well as the impact of colonization and racism.^{2,29,43-45} Moreover, there is a greater proportion of indigenous patients who reside in remote and very remote areas of Australia, where access to and use of health services are more challenging.²⁹

In recognition of a significant deficit in health outcomes of indigenous patients in Australia, on December 20, 2007, the Council of Australian Governments agreed to work with indigenous communities to achieve the target of “closing the gap” on indigenous disadvantage. The Closing the Gap campaign⁴⁶ aims to address gaps in not only the areas of health but also education and employment. One of the three main health goals includes targeting main components of the life expectancy gap and chronic diseases, such as CKD and diabetes mellitus. Some of the listed targets to improve outcomes in this area include attainment of blood pressure targets in patients with type 2 diabetes mellitus and stabilization of all-cause incidence of ESKF within 5–10 years.⁴⁶ In order to help support access to appropriate medications, indigenous patients with chronic diseases or at risk of developing chronic diseases, such as CKD, are eligible to access the Closing the Gap Pharmaceutical Benefits Scheme copayment measure to help access most prescription medications at a lower price or free of charge, depending on eligibility criteria (<http://www.humanservices.gov.au/>). Following implementation of the scheme, improvements in health care access and a declining trend in indigenous mortality rates from chronic diseases have been observed.⁴⁷ However, significant discrepancies between Aboriginal and Torres Strait Island people and the general population remain. These include much higher rates of hospitalization for care involving dialysis (age-standardized rate at ⁴⁵³ per ¹⁰⁰⁰ compared with ⁴² per ¹⁰⁰⁰), with dialysis responsible for ⁴⁴% of all hospitalizations for indigenous Australians. The rate of burden due to CKD for indigenous Australians was eight times that for nonindigenous Australians. Although age-standardized death rates for kidney disease among indigenous Australians had declined, the gap between indigenous and nonindigenous Australians has not changed significantly.⁴⁷

Also, as discussed earlier, there are considerable inequities for Aboriginal and Torres Strait Island people accessing transplantation. To address this, numerous specific initiatives commenced to try to address these inequalities of access. For example, the National Indigenous Kidney Transplantation Taskforce^{28,29} was established to change systems to achieve equitable access to kidney transplantation. A myriad of culturally specific recommendations, along with developing an Aboriginal and Torres Strait Islander kidney health workforce and building effective change-enabling structures to improve the delivery of culturally safe care, have been published.^{28,29}

DETECTION AND MONITORING OF KIDNEY DISEASE

One of the main predictors of poor patient outcomes in those with ESKF has included late referral, defined as less

than 3 months between referral and KRT commencement.¹⁴ Although there are cases for which prevention or early recognition may not be possible, most CKD cases should have been able to be recognized in primary care and reviewed by a nephrologist before imminently requiring KRT. In appreciation of the importance of early recognition of CKD, several initiatives have been developed and evolved over the years in Australia. These have focused on empowering health care providers to recognize risk factors for CKD to trigger screening, optimally manage risk factors to retard progression of disease, and know when a specialist referral is indicated. These include the “Kidney Health Check” through Kidney Health Australia⁴⁸ and the Primary Care Education Advisory Committee for Kidney Health Australia to increase awareness, as well as implementation of best practice detection and management of CKD (www.kcat.org.au).⁴⁹ Other strategies to improve quality of care delivery in primary care settings include development of customized software programs to integrate with primary care electronic health care records. This allows real-time promotion of detection and management of CKD according to best practice recommendations (eMAP:CKD)⁵⁰ and provision of decision support tools for health care providers (Health Tracker, George Institute).⁵¹

Although clear guidelines exist on indications for referral to a nephrologist in Australia (e.g., eGFR <30 mL/min/1.73 m²),⁵² some remote living patients are unable to be reviewed easily by nephrologists. In areas where telehealth facilities are available, nephrologists from the nearest renal units have managed selected patients with CKD via telehealth with local nursing support and coordination.⁵³

HEALTH INFORMATION SYSTEMS AND STATISTICS

Information on patients with ESKF receiving KRT is well described through the ANZDATA Registry, which collects a wide range of statistics relating to the outcomes of treatment of those with ESKF from all renal units in Australia and NZ (www.anzdata.org.au). Data relating to CKD are also collected through the Australian Institute of Health and Welfare (AIHW) (www.aihw.gov.au) and Australian Health Survey from the Australian Bureau of Statistics (<http://www.abs.gov.au/australianhealthsurvey>).⁵⁴ To further enhance understanding of disease and outcomes, several newer registries have been developed over the past decade, including statewide collaborative multidisciplinary research and practice programs in various states across Australia, such as CKD.QLD (Chronic Kidney Disease in Queensland, <https://cre-ckd.centre.uq.edu.au/CKD.QLD>), the Registry of Kidney Diseases (Victoria), and for specific disease groups, such as for glomerulonephritis (QLD. GN Registry).

AOTEAROA NEW ZEALAND

Aotearoa/NZ is situated 2000 km southeast of Australia. Comprising two large islands and several smaller islands, NZ was first settled following migration across the Pacific Ocean in the late 13th century by Māori, the indigenous population. Māori settled throughout Aotearoa, living within iwi (tribes), hapū (subtribes), and whānau (extended families). Abel Tasman was the first European explorer known to have reached NZ in 1642. Progressive settlement by European settlers occurred in the 18th century for whaling

and sealing. Progressive colonization in the 19th century by the British Crown resulted in the signing of the Treaty of Waitangi in 1840 between the Crown and iwi. After World War II, extensive migration to New Zealand by people in the Pacific occurred. Currently, of the 5.12 million people living in NZ, 70.2% identify as NZ European, 16.5% as Māori, 8.1% as Pacific people, and 15.1% as Asian and 1.5% other.⁵⁵

ACCESS TO HEALTH CARE

The NZ health care system is predominantly publicly funded by taxes. Twenty District Health Boards (DHBs) have recently been amalgamated into a single health authority—Te Whatu Ora, with the proposed aim to improve health equity across the nation. Te Whatu Ora regional entities are responsible for funding health care services to their catchment population within a geographic region, providing acute care services, and promoting the health of the population. All public hospital and specialist outpatient services within the public system are free, while attendance in primary care incurs some payment, based on income. KRT (dialysis or kidney transplantation) is provided nearly entirely within the public health care system. Renal specialists provide dialysis and transplantation care through hospital-based remuneration and are not incentivized to provide specific dialysis treatment modalities. Citizens or permanent residents of NZ and dependencies (Niue, the Cook Islands, and Tokelau) receive treatment for ESKF within the tax-funded public health system in New Zealand. Auckland, the largest city in New Zealand, has an estimated population of 1.6 million and is also home to the largest population of Pacific people (approximately 240,000) of any city in the world.

BURDEN OF CHRONIC KIDNEY DISEASE

In Aotearoa/New Zealand, the true proportion of the population who have CKD is unknown, as census or systematic observational data have not been reported. The prevalence of CKD, defined as an eGFR <60 mL/min/1.73 m² or the presence of albuminuria >3 mg/mol (A2-3), has been estimated in the Otago Southland region⁸⁰ (a population of just under 400,000) at 11.8% consistent with prevalence rates in a comparable Australian population.⁸¹ Increasing age, female sex, ethnic groups (Māori and Pacific people), social deprivation, and diabetes mellitus were associated with a higher risk of CKD.⁵⁶ CKD was more prevalent in Māori (odds ratio 1.56, CI 1.45–1.69) and Pacific people (odds ratio 2.62, CI 2.28–3.01) even after adjustment for confounding factors including the presence of diabetes mellitus when compared with the NZ European population.⁵⁶ These data emphasize the importance of nondiabetic kidney disease in Māori and Pacific people despite diabetic nephropathy being the most common cause of ESKF in these groups. Previous studies have demonstrated higher rates of glomerular and hypertensive kidney disease in Māori and Pacific people.^{57,58}

More recently, data from the two largest primary health care providers in South Auckland with 25,132 registered patients, of whom 37.5% were NZ European, 29.7% Samoan, 12.2% Māori, and the rest either other Pacific Island groups or Asian, demonstrated an overall prevalence of CKD of 13.0% (Table 79.1).⁵⁹ However, the prevalence of CKD in the Samoan cohort was 17.8%, Māori 10.4%, and NZ

European 7.1%, with the other Pacific Island groups having a prevalence between 15.7% and 21.8% for CKD. However, only 47.5% of this population had actually been tested for CKD despite the known increased risk of CKD within Māori and Pacific communities. The prevalence of CKD among those actually tested, using the KDIGO criteria for CKD,⁶⁰ rose to 36.3% for Samoans, with similar high percentages for other Pacific Island groups and Māori at 24.1%, and NZ European at 15.6% (see Table 79.1).⁵⁹ The true prevalence probably lies between these two sets of values. It is important to note that individuals registered by primary health care providers represents nearly the entire population. To access primary health care in New Zealand, an individual needs to be registered with a primary health care provider.

Approximately 700 NZ patients (including 28 children and adolescents) commenced KRT and 187 patients received a kidney transplant in 2021 while 472 patients are actively on the waiting list for kidney transplantation.⁶¹ The incidence of new patients starting KRT in NZ was 137 pmp in 2021; although the incidence has largely plateaued in the past 2 years, it reflects an ongoing increase over time with currently 5452 patients (1054 pmp) receiving KRT consisting of 2297 individuals with a transplant and 3155 on dialysis.⁶¹ There is considerable regional variation in the incidence of treated ESKF across the country with higher rates in Northern regions, excluding higher-income areas of Auckland.

Diabetes is the most frequent cause of treated ESKF in Aotearoa/New Zealand, with 54% of patients treated with dialysis and 22% of kidney transplant recipients having a diagnosis of diabetes.⁶¹ Risk factors for CKD frequently coexist including socioeconomic deprivation, obesity, dyslipidemia, and hypertension.^{62,63}

The annual health expenditure in Aotearoa/NZ has risen on average 5.1% compared with annual population growth of 1.3% between 1990 and 2010. The total health expenditure was NZ\$30 billion in 2022, and in 2022 it accounted for 11.2% of real GDP, like many peer nations in the Organization for Economic Co-operation and Development.^{64,65} The estimated health costs attributable to CKD in NZ have been estimated, comparing a cohort commencing dialysis in 2014/2015 with a cohort receiving a kidney transplant in 2014/2015. The first year of dialysis was associated with total patient costs of NZ\$150,878 of which the majority was dialysis costs. Projected costs over the life expectancy of each cohort estimated that dialysis will cost >NZ\$1 million compared with NZ\$0.5 million for a transplant recipient. In addition, the value of improved quality of life from transplantation could be up to NZ\$0.5 million. In addition, loss of productivity for those on dialysis was estimated to be NZ\$47,026 per person per year.^{64,65} Recent studies have estimated the direct costs of diabetes in NZ to account for NZ\$2.1 billion (0.67% of NZ GDP) with the bulk of the costs being direct public health system costs, but economic costs to the individuals and their families are also substantial (approximately half of the estimated total costs).⁶⁶ The provision of dialysis is estimated to cost 1% of New Zealand's health expenditure.⁶⁷

INEQUITY BASED ON ETHNICITY

The incidence of treated ESKF in NZ is not equitable, with an incidence of 88 pmp for New Zealand European patients,

Table 79.1 Prevalence of Chronic Kidney Disease in Patients Attending Two Pacific Island Health Providers in Auckland, New Zealand Between January 1, 2015 and December 31, 2017⁵⁹

	Total Sample (Column %)	Number of People Not Tested for CKD (% Out of Total)	Number of People Tested for CKD (% Out of Total)	Number with CKD	Prevalence of CKD Out of Total Sample [95% CI]	Prevalence of CKD Out of Those Tested [95% CI]
All patients	25,127 (100.0)	13 186 (52.5)	11,941 (47.5)	3279	13.0 [12.6, 13.4]	27.5 [26.7, 28.3]
CKD Grades						
No CKD when tested	8662 (34.5)					
CKD 1				491	2.0 [1.8, 2.1]	4.1 [3.8, 4.5]
CKD 2				1103	4.4 [4.1, 4.7]	9.2 [8.7, 9.8]
CKD 3				1483	5.9 [5.6, 6.2]	12.4 [11.8, 13.0]
CKD 4				112	0.4 [0.4, 0.5]	0.9 [0.8, 1.1]
CKD 5				90	0.4 [0.3, 0.4]	0.8 [0.6, 0.9]
Not tested		13,186 (52.5)				
Age Groups						
15-34	11,910 (47.4)	8860 (74.4)	3050 (25.6)	347	2.9 [2.6, 3.2]	11.4 [10.3, 12.6]
35-44	4042 (16.1)	2059 (50.9)	1983 (49.1)	322	8.0 [7.1, 8.8]	16.2 [14.6, 17.9]
45-54	3728 (14.8)	1184 (31.8)	2544 (68.2)	639	17.1 [15.9, 18.4]	25.1 [23.4, 26.9]
55-64	2960 (11.8)	685 (23.1)	2275 (76.9)	771	26.0 [24.5, 27.7]	33.9 [31.9, 35.9]
65-74	1627 (6.5)	270 (16.6)	1357 (83.4)	706	43.4 [41.0, 45.8]	52.0 [49.3, 54.7]
75-84	687 (2.7)	93 (13.5)	594 (86.5)	388	56.5 [52.7, 60.2]	65.3 [61.3, 69.1]
>85	173 (0.7)	35 (20.2)	138 (79.8)	106	61.3 [53.6, 68.6]	76.8 [68.9, 83.6]
Sex						
Male	11,783 (46.9)	6173 (52.4)	5610 (47.6)	1516	12.8 [12.3, 13.5]	27.0 [25.9, 28.2]
Female	13,344 (53.1)	7013 (52.6)	6331 (47.4)	1763	13.2 [12.6, 13.8]	27.8 [26.7, 29.0]
Ethnicity						
Non-Māori-Non-Pacific	9415 (37.5)	5112 (54.3)	4303 (45.7)	672	7.1 [6.6, 7.7]	15.6 [14.5, 16.7]
Samoan	7451 (29.7)	3806 (51.1)	3645 (48.9)	1324	17.8 [16.9, 18.7]	36.3 [34.8, 37.9]
Tongan	1983 (7.9)	977 (49.3)	1006 (50.7)	356	18.0 [16.3, 19.7]	35.4 [32.4, 38.4]
Cook Island Maori	698 (2.8)	347 (49.7)	351 (50.3)	120	17.2 [14.5, 20.2]	34.2 [29.2, 39.4]
Fijian	700 (2.8)	321 (45.9)	379 (54.1)	110	15.7 [13.1, 18.6]	29.0 [24.5, 33.9]
Niuean	522 (2.1)	233 (44.6)	289 (55.4)	114	21.8 [18.4, 25.6]	39.4 [33.8, 45.3]
Tokelauan	84 (0.3)	44 (52.4)	40 (47.6)	13	15.5 [8.5, 25.0]	32.5 [18.6, 49.1]
Other Pacific Island	1192 (4.7)	604 (50.7)	588 (49.3)	247	20.7 [18.5, 23.1]	42.0 [38.0, 46.1]
NZ Maori	3062 (12.2)	1734 (56.6)	1328 (43.4)	320	10.4 [9.4, 11.6]	24.1 [21.8, 26.5]
Not recorded	20 (0.1)	8 (40.0)	12 (60.0)	3	15.0 [3.2, 37.8]	25.0 [5.5, 57.2]

Modified with permission from Table 3: Tafuna'i M, Turner RM, Richards R, Sopoaga F, Walker R. The prevalence of chronic kidney disease in Samoans living in Auckland New Zealand. *Nephrology*. 2022;27(3):248–259.

compared with incidences of 209 pmp for Māori and 398 pmp for Pacific patients, giving a 3.5-fold difference in the incidence of treated ESKF (Fig. 79.5).^{61,68} These rates continue to increase, especially for Pacific people (Fig. 79.6).⁶¹ There is considerable regional variation in the incidence of treated ESKF across the country with higher rates in Northern regions, excluding higher-income areas of Auckland. Notably, the age of onset for diabetes diagnosis is 10 years younger among non-European ethnicities in NZ, in parallel with the premature onset of treated ESKF at younger age groups for Māori and Pacific patients.⁶⁸

The prevalence of individuals treated for ESKF in NZ varies over threefold by ethnicity, with a prevalence of 767 pmp for NZ European patients, 1453 pmp for Māori, and 2656 pmp

for Pacific patients.⁶⁸ The most common cause of treated ESKF for incident patients in NZ varied due to ethnicity with diabetic kidney disease accounting for 66% in Māori, 63% in Pacific, and 28% in NZ European; glomerulonephritis accounting for 13% in Māori, 12% in Pacific, and 25% in NZ European; hypertension accounted for 13% in NZ European, 5% in Māori, and 4% in Pacific; and polycystic kidney disease accounted for 8% in NZ European and only 1% in Māori and Pacific.⁶⁸ Overall, the incidence rates for Māori and Pacific are markedly higher than those for NZ Europeans with the inequity likely confounded and underestimated by the younger age at presentation for Māori and Pacific people.⁶⁸

There is marked inequity in kidney transplantation rates in NZ based on ethnicity. Despite an age-standardized risk

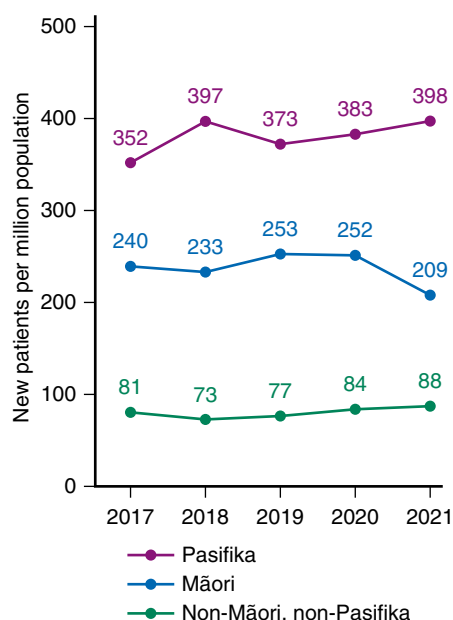


Fig. 79.5 Incidence of kidney replacement therapy by ethnicity—Aotearoa, New Zealand 2017–2020. (From ANZDATA Registry. 45th report. Chapter 9. Kidney failure in Aotearoa New Zealand. Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia. <<http://www.anzdata.org.au>>; 2022.)

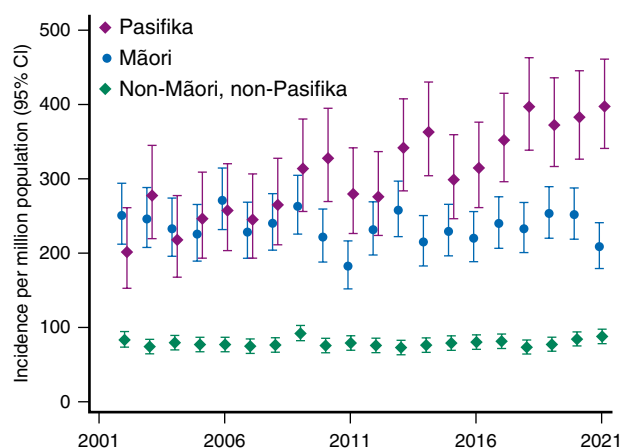


Fig. 79.6 Incidence of kidney replacement therapy—Aotearoa, New Zealand. (From ANZDATA Registry. 45th report. Chapter 9. Kidney failure in Aotearoa New Zealand. Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia. <<http://www.anzdata.org.au>>; 2022.)

of ESKF for Māori and Pacific patients, approximately 2 to 10 times that for NZ European patients, the rate of kidney transplantation is markedly lower.⁶⁸

In 2021, 37 Māori patients received a kidney transplant (42 pmp) while 26 Pacific patients received a kidney transplant (64 pmp), compared with 120 NZ Europeans (32 pmp). This compares with dialysis incidences of 216 pmp and 272 pmp for those populations, respectively.⁶⁸ In addition, recent increases in preemptive kidney transplantation appear to be favoring NZ European patients, whereas Māori and Pacific are less likely to receive a preemptive transplant when compared with NZ Europeans. This phenomenon of increased disparity in transplantation rates during periods

of improved access through better service delivery have been similarly observed in Australia, Canada, and the United States. In addition, there is considerable variation between NZ renal centers in the rate of transplantation, varying between 9 pmp and 46 pmp, which in part reflects geographic isolation.⁶⁸

Māori patients with CKD have experienced marginalization within the NZ health care system due to a number of factors including inequities in the provision of early detection and preventive care, rurality, delayed referral to specialist care, the impact of systemic racism and colonialization, and a lack of awareness by clinicians of important cultural aspects related to health and the involvement of family (whanau) in decision making.⁶⁹ In addition, there is often a multigenerational fear of dialysis, which needs to be overcome by the early involvement of whanau and peers in identifying CKD and aligning decision making with patient priorities.^{69,70} In recognition of these inequalities, members of Caring for Australians and New Zealanders with kidney impairment (CARI) developed clinical practice guidelines for management of CKD for Māori in Aotearoa NZ.¹ They highlight the need for cultural safety; the need to address social determinants of risk factors for kidney health, such as justice, housing, education, and poverty; the provision of whanau-based care (family) within an indigenous health framework including building knowledge about CKD and health issues that contribute to it using whanau-focused and culturally safe strategies.¹

DIALYSIS

The incidence of treated ESKF in NZ is similar to that of neighboring Australia (127 pmp in 2021). The incidence of dialysis varies markedly by nephrology center catchment area with the highest rate for Counties Manukau (including South Auckland; 611 pmp) and Waikato (496 pmp) and lowest for Southern (45 pmp) and Canterbury (78 pmp) regions, with differences between regions related in part to ethnicity and socioeconomic opportunity.⁶¹

Overall, 3155 people were treated with dialysis in NZ in 2021, accounting for a prevalence of 610 pmp. Together with kidney transplantation (444 pmp), the overall prevalence of KRT for Aotearoa/NZ in 2021 was 1054 pmp, which is nearly identical to neighboring Australia (947 pmp) and comparable with the average of the nation-states of the European Union (1090 pmp).⁶¹

Older age groups have the highest incidence of KRT with the highest incidence among people in the 65- to 74-year-old age group. However, for Māori and Pacific people, the highest incidence is in the 45- to 54-year-old age group, which in turn has a far greater impact on both the individual and their whanau (family) because they are often the key income earners or have the major role in family care.⁶⁸ Of the 28 children and young adults (0 to 25 years old) commencing KRT in 2021, 2 received a preemptive transplant, 8 commenced PD, and 18 commenced hemodialysis. The proportion of adults older than 85 years of age starting dialysis remains low.^{14,61}

Overall, most prevalent patients with treated ESKF in NZ are treated with dialysis (61.2%; Fig. 79.7).⁶¹ NZ has the highest global rate of home-based dialysis therapy (38% of those on dialysis). In 2021, 12.0% of patients treated with dialysis in NZ did home hemodialysis, 26% did PD (CAPD

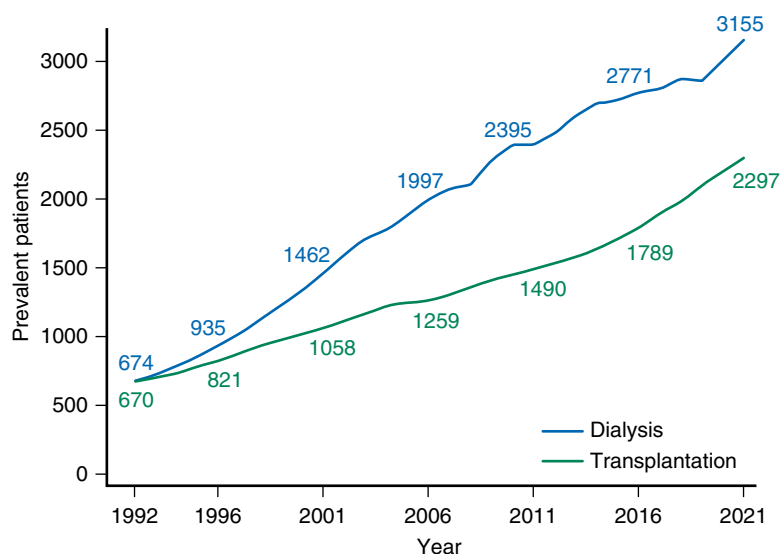


Fig. 79.7 Prevalence of dialysis and transplantation in Aotearoa New Zealand. (From ANZDATA Registry. 45th report. Chapter 9. Kidney failure in Aotearoa New Zealand. Australia and New Zealand Dialysis and Transplant Registry, Adelaide, Australia. <<http://www.anzdata.org.au>>; 2022.)

or APD), and 62% received faculty (hospital or satellite) hemodialysis. There is marked regional variation in home hemodialysis uptake, ranging from 12% of all dialysis patients treated at the Capital and Coast District region to 58% at the Southern region. Regional variation is likely related to the different clinical approaches to the provision of home dialysis (two centers, Canterbury and Southern), maintain an extensive home dialysis program,⁶¹ remoteness, comorbidity, and socioeconomic factors. Home-based hemodialysis among children is rare.

In NZ, hemodialysis is accessed in three principal settings: facility (hospital or satellite), community house, or home.⁷¹ Community house hemodialysis involves patients performing their hemodialysis independently without direct nursing or medical supervision in an unstaffed nonmedical community homelike setting and is used in two DHB regions (Counties Manukau and Hawke's Bay), serving approximately 650,000 people. In a 10-year analysis between 2000 and 2010, 113 patients commenced dialysis in a community house setting.⁷¹

Median patient survival on hemodialysis in NZ varies with modality, age, and ethnicity.⁶¹ The median survival for those aged 25 to 44 years is 8 years, 45 to 64 years is 5.5 years, 65 to 74 years is 3.7 years, 75 to 84 years is 2.8 years, and 85+ years is 2.2 years. There is a 52% lower risk of mortality with home dialysis during the first 3 years of dialysis and beyond, although this difference may not be present for Pacific patients.^{61,71} The median survival for PD in NZ is 3.92 years (95% CI: 2.11–6.34), although it varies substantially by age, comorbidity, and time.^{20,61,72}

The costs of dialysis are not publicly available. Estimations of dialysis and transplant treatment costs are available from the Auckland Regional Renal Report in 2002–2004. In that report, annual costs (including routine dialysis treatments, hospitalizations, physician assessment not related to dialysis, and transportation) were estimated at US\$44,053 for hospital hemodialysis, US\$32,995 for

satellite hemodialysis, US\$25,078 for continuous APD, and US\$23,003 for home hemodialysis. Costs of APD are not available.

TRANSPLANTATION

Three centers perform kidney transplantation in NZ: Auckland, Wellington, and Christchurch. Simultaneous kidney and pancreas transplants are performed in Auckland, as are combined solid organ transplants. In 2021, there were 187 kidney transplants performed in NZ, of which 87 were from live donors. This represents a rate of 36 pmp. Of the recipients, Māori made up just 20% despite the much higher proportion of Māori with ESKF.^{25,61} The number of transplant operations each year has continued to increase over the past 5 years and is likely to be related to quality improvement initiatives arising from government funding over the previous decade.⁷³ Other contributing factors include the NZ kidney exchange program and ABO-incompatible kidney transplantation. Of note, transplant rates in NZ were not dramatically affected by the COVID-19 pandemic. Currently, there are ²²⁹⁷ people with a functioning graft in NZ.^{25,61}

As is observed in Australia, the most frequent cause of graft loss is death with a functioning transplant (442 patients during 2012–2021; 51.6%), followed by chronic allograft nephropathy (193 patients; 22.5%). The most common causes of death for transplant recipients are cancer (29.8% of all deaths), with nonmelanoma skin cancer and lymphoma the major two malignancies, and CVD (25.5% of all deaths).^{25,61}

ACUTE KIDNEY INJURY

Few systematic data exist for the incidence and outcomes related to AKI in Aotearoa/NZ in adults. Limited data suggest that the crude hospital mortality for patients admitted to intensive care with AKI (42.7%) is substantially higher than for those patients without AKI (13.4%), although in the

decade through 2006, mortality decreased among patients with AKI.⁷⁴

For children, the annual incidence of AKI between 2001 and 2006 was 4.0 per 100,000 children younger than 15 years of age, most commonly associated with cardiac surgery (58%), hemolytic uremic syndrome (17%), and sepsis (13%).⁷⁵ Survival of AKI to hospital discharge for children was 89%, with evidence of ongoing kidney impairment in 40%. Māori and Pacific children are treated for AKI more often in part related to the high rates of rheumatic heart disease and poststreptococcal glomerulonephritis in both Māori and Pacific children compared with NZ European children.⁷⁵

HEALTH FINANCE

CKD and AKI are managed collaboratively across the health sector in primary, secondary, and tertiary care. At present, hospital-based dialysis services are provided by 11 secondary or tertiary-level hospitals to their own regions. CKD is most identified within primary care settings, and referral is made to nephrology services for virtual or in-person consultation.

Expenditure on ESKF care is about 1% of total health expenditure. ESKF treatment is nearly entirely funded through the publicly funded health system. All inpatient and outpatient secondary-level health care services including hospital-based care are free to patients, as are pharmaceuticals prescribed within secondary services. Primary care services are subsidized, although patients may be required to pay for services above any subsidy they receive. Similarly, prescriptions in the primary care setting accrue direct costs to the patient.

Regional centers based around a tertiary hospital provider are provided a budget, weighted according to the services they provide and the demographic and socioeconomic characteristics of their population. Budgetary constraints occur, such that centers are incentivized to manage funds within their allocation including dialysis and transplantation care. This has led to inequities in accessing timely and appropriate kidney care, especially as it related to those who live rurally, which includes a significant number of Māori.^{69,76} In particular, there is little in the way of subsidies for travel to a regional center, which further adds to the inequities of accessing care, especially for Māori who frequently cannot afford the costs of travel or the loss of income associated with having to spend considerable time accessing health care from remote settings. Qualitative research has documented the considerable social disruption arising from dialysis treatment in NZ because of living remotely from dialysis and transplantation care.⁷⁶ This has also contributed to inequities in the delivery of health care, especially implementation of kidney disease management.

DETECTION AND MONITORING OF KIDNEY DISEASE

Late assessment by a specialist nephrologist (within 3 months of requiring KRT) is associated with poorer clinical outcomes and reduces the likelihood of preemptive transplantation.⁷⁷ In NZ, approximately 10% of patients experience late assessment, although this proportion is decreasing,⁶¹ with rates for Māori (12%) and Pacific people (9%) compared with NZ European (10%).

There have been some nationwide programs in NZ to guide primary and secondary care practitioners in assessment and referral of patients with CKD. These include the Best Practice Advocacy Centre (bpac^{nz}) detailed review of CKD management for general practitioners (GPs) (<https://bpac.org.nz/2022/ckd.aspx>),⁷⁸ along with Health Pathways, which are principally designed to support primary care clinicians to manage patients in the community and guide referral practices for specialist assessment (www.healthpathwayscommunity.org). Decision aids for patients and clinicians making choices about treatment for ESKF have been developed by Kidney Health NZ, an established patient advocacy group.⁷⁹

A randomized clinical trial of community-based visits by nurse-led health care assistants for medication adherence and algorithm-guided blood pressure care in Māori and Pacific patients led to greater use of medications, lower blood pressure, and reductions in proteinuria, suggesting an effective model of primary care for CKD.⁸⁰ Unfortunately, when funding finished for this project, the benefits were quickly lost. Challenges to wider adoption of newer primary care practices to prevent ESKF include limited resourcing for nursing, availability of secondary care services to upskill primary care clinicians, and adequate protocols to integrate medical management across primary and secondary care. A recent systematic review of studies examining the effectiveness of chronic disease management programs of indigenous people in Australia, NZ, and Canada demonstrated that effective and acceptable programs were those that were integrated in existing health services, were nurse led, provided intensive follow-up, had governance structures supporting local leadership, included robust clinical systems for communication, and involved indigenous health workers as central to practice.⁸¹

HEALTH INFORMATION SYSTEMS AND STATISTICS

NZ has a National Health Index that assigns a unique number to each person in NZ who receives health care. This includes information on name, address, date and place of birth, gender, resident and citizen status, ethnicity, and (if appropriate) date of death.⁸² The National Health Index number can be linked to other national and international health information repositories. An important aspect of health information used to drive health care improvement is the accurate recording of ethnicity data, especially for Māori and identifying health inequalities. Specific protocols for the collection, recording, and output of ethnicity data are described for the Health and Disability Sector, with a specific focus to improve health outcomes and reduce health inequity.⁸³ Key requirements include 1. ethnicity as self-identification, 2. ethnicity is context specific and can change, 3. more than one ethnicity can be recorded, and 4. a standardized approach increases the accuracy of ethnicity data for decision making. It is likely, due to increasing availability of large data sets, routine capture of diagnostic codes, and the National Health Index, that additional census data for kidney diseases will become available in the future.

As well as a strong focus on addressing the health inequities that Māori face, there is also heightened awareness of the

similar health issues faced by Pacific people living in NZ, as well as in Pacific Islanders. The next sections highlight the impact of kidney disease in the Pacific Islands.

SAMOA

Samoa is made up of six islands and is a member of the Commonwealth of Nations. From the end of World War I until 1962, when Samoa attained independence, it was under the control of NZ. The population of Samoa is around 200,000 people. Like other island nations in the Pacific, noncommunicable diseases including diabetes, hypertension, CVD, and CKD are highly prevalent in Samoa. CKD is listed within the top 10 causes of deaths and disability (DALY) in the country (ranked 4th in 2019).⁸⁴ High body mass index (BMI), hyperglycemia, hypertension, and kidney dysfunction are major risk factors contributing to disability-adjusted life-years. In 2019, the National Kidney Foundation of Samoa (NKFS) undertook an extensive community screening program to detect the estimated prevalence of CKD and its associated risk factors in Samoa.⁸⁵ There were 11 urban and rural sites on Upolu (the largest island) and 4 rural sites on the island of Savaii. A total of 1163 adults were screened for CKD using point-of-care serum creatinine determinations and eGFR using the CKD-EPI formula along with dipstick urinalysis. Data were captured for risk factors of obesity, diabetes (using point-of-care HbA1c determination), and blood pressure. The presence of CKD (grades 1–5) was 44.5% (95% CI 41.6%–47.4%) with a mean eGFR of 54.3 ± 20.8 mL/min/1.73 m². Rates were similar between males and females. This equates to an age-standardized total CKD prevalence of 31.8%, which is almost three times the global prevalence of 11% to 13%.^{9,85} Ninety-seven individuals had dipstick-positive proteinuria, and 109 (9.4%) had dipstick-positive hematuria. Obesity (BMI >32) was 66.3%, diabetes was 30.8%, and hypertension was 54.3%.⁸⁵ This high prevalence of CKD suggests that changes associated with colonization, urbanization, and westernization are now critical in Samoa, as well as other Pacific Island countries.⁸⁶ The finding that 9.4% of the individuals tested had microhematuria raises the possibility of undiagnosed glomerulonephritis contributing to the increased prevalence of CKD. Unfortunately, Samoa's limited health resources prevent any further analyses of causes for CKD. Previous studies from NZ have demonstrated that Pacific people have higher rates of ESKF due to glomerulonephritis when compared with Māori or NZ Europeans.⁸⁷

Tafuna'i and colleagues⁸⁸ have reported on the epidemiology of kidney failure patients receiving hemodialysis in Samoa. The NKFS has provided hemodialysis since 2005. According to the NKFS database, 393 individuals have received dialysis between 2005 and 2019. Diabetic nephropathy (64.9%) was the predominant cause of ESKF. Forty-three percent were women, and the mean age at commencement of dialysis was 54.9 years. On the basis of the number of treated dialysis patients, this equates to a crude mean incidence rate of 224 patients pmp and a crude prevalence rate of 629 patients pmp.⁸⁸ Although the median survival on dialysis was 2.9 years, 33.5% of people commencing dialysis die within the first 12 months of starting dialysis.⁸⁸

There are currently two dialysis centers located in two main islands (Upolu and Savaii), with the NKFS just opened a new upgraded dialysis center with 60 stations at Moto'otua Apia (Upolu), the capital of Samoa. The smaller unit based on Savaii has 11 dialysis chairs. PD is not available. Although the NKFS is a state-owned organization separate from the Ministry of Health and the National Hospital based in Apia, individuals are referred to the NKFS via the hospital of private medical practitioners. The government partially funds the cost of dialysis; individuals must pay US\$3.70 per dialysis session. Low incomes and rurality are significant limitations to accessing dialysis, which means the crude mean incidence of ESKF mentioned earlier probably underestimates the true incidence of ESKF in Samoa.

Both the NKFS and Ministry of Health in Samoa have acknowledged the strain imposed by the growing burden of CKD and are currently working on public awareness campaigns and evaluating methods to promote healthier lifestyle measures to lower risk factors for CKD, such as hypertension and diabetes.

Acute dialysis is provided within the Tupua Tamasese Meaole (TTM) Hospital and is separate from the NKFS. A retrospective case series of AKI admissions to TTM in more than a 6-month period in 2020 identified 114 AKI admissions, using KDIGO criteria for diagnosis of AKI.⁸⁹ This equates to a hospital-based incidence of 26.8% per 1000 admissions per 6 months. Most (75% of) cases were community-acquired AKI due to intravascular volume depletion (79%) and sepsis (64%). The majority of patients were in the 50- to 69-year age group with multiple comorbidities including type 2 diabetes mellitus, existing CKD, and/or CVD. Twenty-five percent of those hospitalized for AKI died within a 3-month period (10 in hospitals and 13 at home). Although 27 cases met indications for acute hemodialysis, only 5 of them agreed to proceed with hemodialysis and 3 of those individuals died on dialysis.⁸⁹ Reasons for not accepting dialysis support are complex but certainly have an important cultural overlay including health beliefs and the links to traditional ways of living (fa'aSamoa), as well as traditional healers and attitudes toward dialysis, perhaps linked to a perception that it is the dialysis, rather than the advanced stages of life-threatening illness, that is the cause of death.⁸⁸⁻¹⁰⁰

NEW ZEALAND DEPENDENCIES (COOK ISLANDS, NIUE, TOKELAU)

Cook Islands, Niue, and Tokelau are all self-covering island countries in the South Pacific Ocean in free association with NZ. Cook Islands is the most populous of the group, which comprises 15 islands. Tourism is the country's main industry as it lacks major natural resources (makes up approximately 67.5% of GDP). The Cook Islands NCD risk factors STEPS report from 2013–2015 identified the prevalence of obesity was 68.7%, hypertension 28.5%, and diabetes 23.5%.¹⁰¹ Although Cook Islands is thought to perform better than other Pacific Island nations in relation to health indicators, there has been a 40% to 50% increase in the prevalence of diabetes, followed by an increasing number of CKD patients (expected to more than double in the next 10 years).¹⁰¹ Noncommunicable

diseases such as cardiovascular diseases and CKD are reported to be the leading causes of mortality in the Cook Islands (responsible for 75% of deaths).¹⁰² Using data from the Pacific Island primary health care providers in Auckland, the prevalence of CKD in Cook Island Māori was 17.1% but as high as 34% in those who had actually been tested (see Table 79.1).⁵⁹ There are no nephrology services in the Cook Islands. Patients with ESKF, as NZ citizens, receive dialysis treatment within the NZ tax-funded public health system presently because there is no capacity for KRT within the Cook Islands. Most patients receive hospital-based hemodialysis in NZ, driven by patient preference. This requires relocation to Auckland, staying with extended family, but still incurs considerable personal costs to individuals and their families.

Niue is located 2400 km northeast of NZ and west of Cook Islands, with the population made up predominantly of Polynesians (around 1612 as of November 2016). More than 90% of Niuean people live in NZ. Similarly, about 1400 people are thought to reside in Tokelau, which is another island country located north of the Samoan Islands. Given their small populations with limited data available on health, there is no known information pertaining to the burden of AKI or CKD in either of these two nations. However, using data from Auckland, the prevalence of CKD in Niueans is 39.4% among those who had been tested and 32.5% in Tokelauans who had been tested (see Table 79.1).⁵⁹

PAPUA NEW GUINEA

Papua New Guinea is classified as a low- to middle-income country with close to 90% of the population living in rural areas. Primary health care facilities are the main point of access to the health system, but access to these services faces many geographic, cultural, financial, and systemic challenges.¹⁰³ Many villages can only be reached via air or foot, where >90% of existing roads are unpaved.¹⁰⁴ Over the past 20 years, there has been a deterioration in the availability of primary health care services,¹⁰⁵ with up to 40% of aid posts closing, thereby restricting access to health care. Moreover, despite a predominant public funding base supplemented in some areas with church-run facilities supported by grants from the National Department of Health, many facilities charge a user fee, which creates further barriers to health care for those who cannot afford the service.¹⁰⁶

The exact prevalence of CKD in Papua New Guinea is unknown; however, CKD was ranked as the eighth most common cause of death in 2015.¹⁰⁷ Increase in the burden of CKD is likely to be driven by growth in prevalence of diabetes mellitus, which is thought to affect 14.6% of the adult population in Papua New Guinea.¹⁰⁸ However, access to KRT is limited. The Papua New Guinea Kidney Foundation was set up in 2009 with 2 hemodialysis machines. They now have a dedicated outpatient dialysis facility in Port Moresby, which dialyzes 30 people. They have a smaller second center based at the Mendi General Hospital in the Southern Highlands with two dialysis machines. A further unit is planned for the Kimbe hospital in West New Britain. Currently, dialysis costs are subsidized by funding provided by Kumui Minerals Holdings limited—a state-owned

company whose charter has a commitment to community service.¹⁰⁹ There is no capacity for renal transplantation.

PACIFIC ISLANDS

FRENCH PROTECTORATES (FRENCH POLYNESIA, NEW CALEDONIA, WALLIS, AND FUTUNA)

French Polynesia, New Caledonia, Wallis, and Futuna form the French protectorates within Oceania. French Polynesia is made up of 118 islands scattered over 2000 km of ocean, divided into five groups of islands, with 67 islands that are inhabited.¹⁴⁰ Tahiti is the most populated island of the group and serves as the capital of the collectivity. Although the Government of France directs justice, university education, security, and defense, the local government controls health.¹¹⁰ French Polynesia has a moderately developed economy with modest GDP per capita among countries within Oceania. Health facilities are considered to be of reasonable standard; however, access to medical services including dialysis is restricted in less populated islands. The nephrology service is based at the Taaone Hospital in Papeete, Tahiti. French Polynesia has one of the highest rates of diabetes worldwide with a rate of 26.8%.¹¹¹ In 2022, there were 563 chronic hemodialysis patients and 188 individuals with a functioning kidney transplant. The prevalence rate of ESKF is estimated at 2680 per million inhabitants of French Polynesia, which is double that of mainland France.¹¹² While diabetic nephropathy is a major cause of CKD in this region, there are a significant number of younger Polynesian people (<45 years of age) with ESKF—178 individuals (23.1%), whose etiology of ESKF is not due to diabetes. Of those younger than age 45, only 27% of cases were related to diabetes. Most cases are due to an underlying genetic abnormality (32%) or they are immune mediated because of SLE or IgA nephropathy (14%).¹¹²

Due to a founder effect, two important genetic causes of ESKF in French Polynesia have been identified. The first is related to a tandem duplication within the *COL4A5* gene, leading to high rates of Alport syndrome, which affects 18% of the patients receiving dialysis¹¹³ and is the most prevalent monogenic inherited disorder found within French Polynesia.¹¹³ More recently, a high prevalence of karyomegalic interstitial nephritis has been reported in French Polynesia, with individuals homozygous for the FAN1 c.2120G>A, p.(W707X) variant, with the majority of individuals developing ESKF before the age of 40.¹¹⁴

Like other Pacific Island groups,¹¹⁵ there is a high incidence of lupus nephritis causing CKD, especially in the younger age group of native French Polynesians.¹¹⁶

New Caledonia is another special collectivity of France, located within the Melanesia subregion, formed by several islands including the main island of Grand Terre. It has one of the largest economies in the South Pacific, with GDP per capita similar to that in NZ. According to a retrospective renal biopsy study, which included 202 renal biopsies from 181 patients in New Caledonia, the most prevalent primary glomerular disease was focal segmental glomerulosclerosis and the most prevalent systemic glomerulopathy was amyloidosis. Other common pathologic findings included postinfectious glomerulonephritis, minimal change disease, and diabetic nephropathy.¹¹⁷ In 2009, the most common conditions requiring long-term treatment included diabetes

(9509 cases, 17.6%) and 1198 cases of renal failure (2.2%) requiring treatment. Given the growing burden of diabetes in the community, CKD from diabetic nephropathy is likely to have also increased over time. The local government has endorsed the “Health for All” principle, with provision of public and private health care services (with 2.9 hospital beds available for every 1000 people). Economic growth of New Caledonia and improvement in quality of health care coverage have led to unfettered access to health services by the whole population.¹¹⁸

Access to both hemodialysis and PD is available¹¹⁹ with dialysis facilities on the main island of Grand Terre and a smaller unit on Wallis Island. In 2017 there was a total of 594 dialysis patients in this region with the incidence of newly treated ESKF approximately twice that of France. The prevalence of ESKF in New Caledonia is 2542 pmp in 2017.¹¹⁹ Before 2012, live-donor kidney transplants were carried out in Sydney, Australia or the potential recipient had to relocate to France to await a deceased-donor kidney transplant. Subsequently, a successful deceased-donor retrieval program was set up in Noumea. There were 30 successful transplants as of 2018.¹²⁰

The Territory of the Wallis and Futuna Islands is made up of three main islands with a smaller number of tiny islets. The vast majority of the population is of Polynesian ethnicity, with a small minority of French descent. A large proportion of Wallisians and Futunans live as expatriates in New Caledonia due to limited resources on their homelands, which has led to a decreasing trend in population over time (14,166 in 1996 vs. 11,558 in 2018).¹²² The Health Agency for the Territory of the Wallis and Futuna Islands is a French national public administrative agency with the aim to ensure health protection in the Territory. There is a high prevalence of risk factors for chronic noncommunicable diseases including cigarette smoking (49.1%), obesity (59.5%), hypertension (34.1%), diabetes mellitus (17.1%), and dyslipidemia (16.1%) according to a study that comprised 560 people from the Territory (representing 6.71% of the adult population living in Wallis and Futuna).¹²² In the studied population, moderate CKD, defined as an eGFR of 30 to 59 mL/min/1.73 m², was present in 5.8% of the participants.¹²² As mentioned earlier, there are now dialysis facilities available on Wallis Island.¹¹⁹

US PROTECTORATES (AMERICAN SAMOA, GUAM)

American Samoa is an unincorporated territory of the United States consisting of five main islands. It has a small developing economy with two main sources of income, the U.S. Government and tuna canning, which make up 93% of the economy together.¹²³ The most significant health problems relate to the increase in chronic diseases associated with lifestyle, caused by poor nutrition and physical inactivity, such as hypertension, CVD, and type 2 diabetes mellitus.¹²⁴ According to the American Samoa noncommunicable STEPwise approach to Surveillance (STEPS) survey, conducted between June and August 2004, 74.6% of American Samoans were obese (defined as BMI ≥ 30 kg/m²) and 93.5% were classified as overweight (BMI 25–29.9 kg/m²) or obese, while hypertension and diabetes were present in 34.2% and 47.3% of the adult population, respectively.¹²⁴ More recent data demonstrate CKD as the fourth highest cause of death and DALY with increased BMI,

hyperglycemia, and hypertension the top three risk factors for death and DALY with kidney dysfunction ranked as the sixth highest risk factor.^{125,126} Although the exact burden of CKD is uncertain, it is likely to be significant and growing in the context of an increasing burden of risk factors for CKD.

There is one hemodialysis unit at the Lyndon B. Johnson Tropical Medical Center in Pago Pago, which has 32 dialysis chairs,¹²⁴ where 200 patients with ESKF receive hemodialysis treatment each week.¹²⁴ The number of patients receiving hemodialysis in American Samoa has increased by 33% from 2006 to 2010.¹²⁴ More recently, a private dialysis center has opened in Pago Pago and can treat 50 patients a week.

Guam is another unincorporated and organized territory of the United States. However, unlike American Samoa, Guamanians are American citizens by birth. Guam has a land area of 544 km² (75% the size of Singapore) and is the largest island in Micronesia. The largest ethnic group is the native Chamorros (37.3%), followed by Filipino (26.3%), White (7.1%), and Chuukese (7%) ethnicities. Guam's economy depends primarily on tourism.¹²⁷ Similar to other Pacific Island nations, the most common cause of ESKF is diabetic nephropathy, particularly among Pacific Islanders.¹²⁸ U.S. board-certified medical practitioners serve the people of Guam in all specialties, and health care is available through public and private sectors.

FII

Fiji is an island country in Melanesia formed by >330 islands, of which 110 are permanently inhabited. More than 80% of the population resides on the two main islands: Viti Levu and Vanua Levu. Fiji has one of the most developed economies in the Pacific due to an abundance of natural resources and an active tourism sector. CKD has been identified as the fourth most common cause of death in 2015 (increase by 34.0% since 2005).¹²⁹ A recent STEPS survey has reported 30% of Fiji's adult population as diabetic, and the incidence of stage 5 CKD is estimated to be the highest in the world, at around 680 pmp.¹³⁰ Fiji suffers from an underdeveloped health system, which has led to difficulties to screen and treat early-stage CKD and the urgent need to address noncommunicable diseases including CKD has been recognized by the government. Fiji has three fully trained nephrologists plus a number of locally trained doctors working in nephrology. They are in the process of establishing the Fiji Society of Nephrology and already hold annual scientific meetings with sponsorship from the International Society of Nephrology and Asia Pacific Society of Nephrology.

The nephrology service was set up in collaboration with the International Society of Nephrology sister renal center program based at the Colonial War Memorial Hospital Suva with initial funding for intermittent hemodialysis for AKI. At the start, chronic hemodialysis was not funded, which certainly affected access to KRT. In 2018 the Fiji Government introduced a subsidy for those whose income is less than FJ\$30,000 that covers 50% of the cost of dialysis in the two public hospital units. Before this, all dialysis was on a fee-for-service basis. There are currently six dialysis units in Fiji, two based in public hospitals (Colonial War Memorial Hospital Suva and Labasa Hospital) and four private units dialyzing about 160 people.¹³¹ A retrospective audit in 2012 identified 159 cases of confirmed ESKF,

64% of whom had diabetic nephropathy as the cause of their ESKF. The median age was 57 years (interquartile range 47–65) and a crude age-adjusted rate for ESKF of 938 pmp (95% CI: 804–1072).¹³² Currently there is no PD or transplantation provided, with living-donor kidney transplants being performed in India for those who can afford it.¹³¹ More recently, in conjunction with the George Institute in Sydney Australia, the Fijian dialysis registry was established in 2019 to more accurately document the extent of ESKF.¹³¹ Three of the six units currently provide data to the registry. For its first report, there were 66 patients with a median age of 54 (IQR 44–61); 59% were male, 15 of them were i-Taukei (native Fijian), and 46 of them were Indian. Diabetes (42%) was the predominant cause of ESKF. The majority of patients were dialyzed via double-lumen central venous catheters and most were only dialyzed for 2 sessions of 4 hours a week.¹³²

Acute kidney injury is also an important clinical problem in Fiji, with leptospirosis one of the important causes of AKI and identified as one of the four priority climate-sensitive diseases of major public health concern.¹³³ Outbreaks of leptospirosis have frequently occurred in the setting of floods, which resulted in 576 reported cases and 40 deaths (7% case fatality) in 2012.¹³⁴ Some of the identified risk factors have included male gender, indigenous Fijian status, young adults (15 to 45 years of age), living in rural areas, and abattoir workers, with peak seasons following rainy months and cyclones.¹³⁵ A retrospective study of cases fulfilling KDIGO criteria for AKI identified ¹¹⁰ cases with acute sepsis and/or volume depletion—dehydration in the majority of cases. Fifty-two cases were community-acquired AKI. Twenty-one cases required intermittent hemodialysis, and the overall mortality was ^{43%} including

¹⁶ of those who required dialysis and sepsis contributing to half of the deaths.¹³⁶

THE CARIBBEAN

An estimated 44 million people live in the Caribbean, composed of islands and territories in the subregion of the Americas surrounded by the Caribbean Sea or North Atlantic Ocean and a total land area of 87,258 square miles¹³⁷ (Fig. 79.8). The region is ethnically diverse, with African or black descent, Native, European, and Asian being the predominant ethnicities. There is, however, significant heterogeneity in representation of ethnic groups among countries. Because of the unifying history of European colonization, the most common languages are Spanish, French, and English, followed by Dutch and Portuguese.^{138,139} Although in the same geographic area, countries may be grouped on the basis of cultural, language, and political similarities as Latin American (Dominican Republic, Puerto Rico, and Cuba) or West Indian (e.g., Jamaica, Trinidad, Tobago, and Barbados). Furthermore, some countries, although located in Central or South America, are considered Caribbean countries, such as Belize and Guyana, due to proximity to the coastline and shared political and cultural practices.¹³⁹

Most countries or territories are considered high- and middle-income social developmental index (SDI), with one country (Haiti) being low income.¹⁴⁰ However, like the islands of the Pacific, the Caribbean is considered a geographic region of SIDS. SIDS are a distinct group of countries and states that face unique social, economic, and environmental vulnerabilities due to remote geography, small populations, susceptibility to exogenous economic



Fig. 79.8 Map of the Caribbean region. (From Wikimedia. <https://commons.wikimedia.org/wiki/File:Caribbean_general_map.png> Accessed 06.02.24 and Kmusser, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

shocks, and fragile environmental ecosystems.¹⁴¹ Although less remote from larger economies (e.g., United States and Canada) compared with other SIDS, the Caribbean SIDS have among the highest debt per capita.^{142,143} Due to this lack of fiscal resilience from public debt, the region is particularly prone to the adverse public health effects of climate change and socioeconomic stressors.¹⁷⁴ In terms of climate change, higher rates of illness, economic displacement and death from natural disasters, food and water insecurity, heat stress and dehydration, and infectious disease transmission occur.¹⁴¹⁻¹⁴³

Epidemiologic transition to increased noncommunicable disease burden (diabetes mellitus, hypertension, and obesity) has placed a further strain on the available health care resources.¹⁴⁴ Variability of health care access across countries exists, with those with larger populations and higher SDI (e.g., Cuba, Dominican Republic, and Puerto Rico) having more robust health care infrastructure and personnel.¹⁴⁵ However, economic stagnation has led to the loss of skilled health care personnel to higher-income countries in North America, leading to deficiencies in health care access across the region.

Organization of health care varies by country. In some countries, health care is financed or subsidized through government or national health care organizations, and in others a significant proportion of out-of-pocket payments is required.¹⁴⁵ Despite some heterogeneity in health care delivery and services, demographic, social, and environmental determinants of health, when they are considered in aggregate, the Caribbean hosts a population vulnerable to kidney disease. It has one of the highest burdens of kidney disease in the world, with a high rate of CKD-attributed mortality and disability-adjusted life-years.¹⁴⁶

In this section, we discuss the epidemiology and etiology of AKI and CKD in the Caribbean, diagnostic capacity for kidney disease, and KRT. Finally, we discuss kidney care in two Caribbean countries, Haiti and Jamaica.

EPIDEMIOLOGY OF CHRONIC KIDNEY DISEASE IN THE CARIBBEAN

The prevalence of CKD in the Caribbean was estimated at 9.8% in 2017, with an age-standardized rate of 8591 per 100,000 population, accounting for approximately 11,000 deaths.¹⁴⁶ As of 2019, kidney disease is the 8th leading cause of death and the 10th cause of number of years of life lost in the Americas, representing the largest increase among noncommunicable diseases since 2000.^{146,147} The impact and burden of CKD are likely underestimated due to limited population-based data on prevalence and outcomes in CKD, in part due to lack of surveillance systems, access to electronic records, and registry data in many Caribbean countries.¹⁴⁸⁻¹⁵³ In some Caribbean countries (e.g., Cuba and Puerto Rico), renal surveillance data are integrated into the health care systems and financed through national and regional health care organizations, while others like the West Indian Caribbean's Caribbean Renal registry have voluntary data obtained with a lack of sustainable funding.^{148,149,153,154} Consequently, there are inequities in the reporting of CKD epidemiology among countries in the region, with some countries having limited or no data

and others having population estimates of incident and prevalent CKD.¹⁴⁸⁻¹⁵⁴

CHRONIC KIDNEY DISEASE RISK FACTORS

Diabetes and hypertension are among the leading causes of CKD in the region, paralleling regional increases in obesity, population aging, and urbanization resulting in reduction in physical activity and increased consumption of calorie-dense, high-sodium foods.^{148,151,154-157} Diabetes incidence is increasing in the Caribbean, with estimated prevalence rates ranging from 9% to 18%^{156,157,159} and women and persons of East Indian descent disproportionately affected.^{160,161} Importantly, high rates of noncontrol of diabetes occur in the region (~48%) and with a quarter (~24%) of persons living with diabetes unaware of their diagnosis.^{157,160,161} Of concern, is an alarming increase in childhood obesity (rates of 28% to 35%), which bodes poorly for cardiometabolic health and therefore future kidney disease risk.¹⁶⁸

Hypertension is also a leading cause of CKD in the Caribbean. Rates of hypertension among the non-Latin American Caribbean countries are the highest among the Americas. The estimated prevalence rates of hypertension range from 28% to 55%.¹⁶² In countries such as Jamaica, the Cayman Islands, and Barbados, hypertension-related kidney disease predominates.¹⁶³ Controversy and uncertainty exist regarding the relationship between hypertension and CKD. In the past 10 years, the association of high-risk *APOL1* alleles and hypertensive kidney disease and its impact on incident CKD and its progression has been well described among African American and Afro Caribbean populations resident in North America and Europe.¹⁶³⁻¹⁶⁵ The presence of *APOL1* high-risk alleles confers a 7-11-fold increased risk of incident hypertension-attributed end-stage kidney disease in African Americans.¹⁶⁵ Little is known of the prevalence of *APOL1* high-risk alleles in the region. Furthermore, *APOL1* requires a second hit, such as viral (e.g., HIV) and/or environmental exposure for CKD risk, and/or interaction with other diseases such as systemic lupus erythematosus to increase risk.^{167,169} Exploring the potential interaction between *APOL1* risk alleles and secondary hits is needed in the region.

A high rate of nontraditional risk factors occurs in the Caribbean. Higher rates of lupus nephritis, sickle cell disease, and sickle cell trait are present in some Caribbean countries. CKD unknown, or Mesoamerican nephropathy due to a combination of heat stress, dehydration, and agricultural pollutant exposure, although more common in Central American countries, has been described in a few Caribbean countries.^{166,167} The extent to which these diseases occur warrants a novel area for further study.

DIAGNOSTIC CAPACITY OF CARIBBEAN COUNTRIES

Due to intercountry variability in health care resources and funding, there is variability in diagnostic capacity in the region.¹⁵⁵ Furthermore, much like other middle-income countries in Asia, there is variability in the measurement of creatinine within and between countries. Lack of standardization of estimated glomerular filtration rate (eGFR) measurement and reporting exists.¹⁷⁸ There is significant racial diversity in the region, and the validity and accuracy of newer race-free equations proposed from North America and Europe is unknown and represents

another area for further research.^{179,180} Cystatin-C is not routinely available in most Caribbean countries and must be outsourced at an additional cost to the patients. Indeed, few Caribbean countries have the capacity for in-house laboratory testing for cystatin-C. Renal pathologic services are also limited to larger countries such as Puerto Rico and Jamaica due to lack of nephropathologists and processing capacity for kidney biopsy. Thus in many Caribbean countries, kidney biopsies are outsourced to North America or larger South American countries such as Columbia. The costs of specimen transfer, administrative costs, and processing capacity for tissue diagnosis of renal diseases limit universal access to kidney biopsy in the region. This has implications for diagnosis of glomerular and tubular diseases and adds a layer of uncertainty of the etiology and characteristics of kidney disease.

KIDNEY REPLACEMENT THERAPY RATES, CAPACITY, AND OUTCOMES

The most common form of KRT is in-center hemodialysis. The average rate of treated kidney failure (dialysis or transplantation) varies by country. In the West Indian population with available data on prevalent ESKF, rates vary from 334.6 to 819.5 pmp.¹⁸¹ In Latin American Caribbean countries this ranges from 305 to 2129 pmp.¹⁸² PD is available in variable degrees in many Caribbean countries, with Puerto Rico, Cuba, Trinidad, and Tobago exemplifying well-established PD programs.^{181,183,184} Widespread use of PD has been limited by availability of PD fluid (due to predominant outsourcing) and lack of experience and expertise in PD in the region. Some countries lack the capacity for KRT for ESKF in any form such as Montserrat and Grenada, and this is outsourced to other countries.^{181,185} Kidney transplantation only occurs in some countries, with few having national transplantation programs. These include Cuba, Dominican Republic, Trinidad, Tobago, and Puerto Rico.^{181,182} The development of transplantation programs is under way in many Caribbean countries. In the past decade through collaborations with transplant centers in the United Kingdom and North America, KT has been performed in some of these islands.¹⁸⁷ In Barbados and Jamaica through UK-based charity Transplant Links, ongoing capacity building through site visits, online educational symposia and medical and surgical mentorship has resulted in the performance of live kidney transplants since 2011 in Jamaica and 2016 in Barbados.^{186,187}

EPIDEMIOLOGY AND OUTCOMES OF ACUTE KIDNEY INJURY IN THE CARIBBEAN

Sparse published data exist on AKI incidence and outcomes in the Caribbean. However, a disproportionate degree of the burden and adverse outcomes from AKI and AKD exists in middle-income countries of which the Caribbean countries comprise.¹⁸⁸ Reversible causes of AKI inclusive of dehydration, especially in the context of heat stress; infectious exposures such as dengue virus and leptospirosis, which confer increased AKI risk; pregnancy-related kidney diseases; and nephrotoxin exposures (from herbal medications and fruits) are potential risk factors unique to the Caribbean.^{188,189} Inequities in access to nephrology care, delayed recognition, and dialysis care contribute to higher

morbidity and mortality. On the basis of an observational study of AKI incidence from Latin America, of which few Caribbean countries were enrolled, community-acquired AKI occurred more frequently (in approximately two-thirds) of AKI cases compared with hospitalized AKI, which accounted for 38% of cases.¹⁹⁰ As expected, common causes of community-acquired AKI were dehydration and nephrotoxin exposure, whereas for health care-associated AKI, critical illness (sepsis, cardiac shock) and age were associated risk factors, as with high-income countries.¹⁹⁰ In a single-center study of community-acquired AKI in emergency department visits in Jamaica, dehydration, infection, cardiac failure, and trauma were leading causes.¹⁹¹

Notably, there were significant delays in health care access in community-acquired AKI. On the basis of available data, targeted interventions for early recognition of AKI events, public awareness of AKI risk factors, and infection control strategies are needed.¹⁸⁹⁻¹⁹¹ Data on AKI rates, etiology and outcomes, predictive cost-effective biomarkers in AKI, and implementation studies aimed to reduce the impact of AKI in the Caribbean are needed.

KIDNEY DISEASE IN HAITI

Haiti is one of the largest islands in the Caribbean, with an estimated population of 11.5 million. Recent natural disasters, ongoing poverty, and sociopolitical instability have hampered economic growth, with more than half of Haitians living in extreme poverty and the nation having the lowest gross national per capita in the Western Hemisphere (US\$3310).^{192,193} Noncommunicable disease burden is increasing in Haiti, with hypertension occurring at a younger age and with inadequate blood pressure control (only 13% have controlled hypertension).¹⁹⁴ High levels of stress, obesity, and poor health access contribute to not only lack of hypertension control but also other cardiometabolic risk factors and may explain the high observed cardiovascular mortality. According to a recent national cohort study of 2424 Haitians with creatinine and urine albuminuria testing, the estimated age-standardized prevalence of CKD was 14%, with 3% with eGFR <60 mL/min/1.73 m² and 11% albuminuria. Hypertension, diabetes, and older age (>60 years) increased risk. Notably, more than a quarter of persons with hypertension had CKD and only 12% of persons were on either ACEI or ARB.¹⁹⁵

In terms of pediatric CKD, reported causes are glomerular disease and corticosteroid-resistant nephrotic syndrome. Access to nephrology care is limited; there are seven nephrologists (six adults, one pediatric) in the country, all located in its capital, Port au Prince. In-center hemodialysis is the predominant form of renal replacement therapy, but many persons with ESKF lack access to this therapy due to location (most centers are in Port au Prince) and costs of dialysis. PD and kidney transplantation are not locally available.¹⁹³

In terms of AKI, limitations in health care access in part fueled by low physician-to-patient ratios (<0.5 physicians to 1000 population), environmental vulnerabilities, and lack of laboratory access contribute to adverse outcomes.¹⁹³ In a series of 194 AKI admissions at a 300-bed center in Haiti, heart failure, glomerulopathy, sepsis, and eclampsia were common AKI etiologies. Through training with North

American sister centers, local medical residents performed hemodialysis (due to lack of HD nurses), bedside ultrasound, and kidney biopsy. One-fifth (19%) of these patients received dialysis and 84% survived. Importantly, these services were given free of cost to patients.¹⁹⁶

KIDNEY DISEASE IN JAMAICA

Haiti's neighbor, Jamaica, is the largest Anglo Caribbean country, with a population of 2.9 million. Despite fiscal growth in the past decade, the island is an upper-middle-income SDI, with the lowest GDP per capita among English Caribbean islands.¹⁹⁷

Like Haiti, CVD is the leading cause of death. One in three Jamaican adults have hypertension, and 12% have diabetes.¹⁹⁷ CKD is the fourth leading cause of death and the fifth leading cause of death and disability. There are 16 (12 adult, 4 pediatric) nephrologists in the island (mean 5.5 per 100,000 population).¹⁹⁷ There are limited published population-based data on CKD prevalence. On the basis of the Jamaican arm of the renal registry, hypertension is the most common cause of ESKF, followed by diabetes and lupus nephritis; however, given the voluntary nature of this effort, ESKF rates are underestimated. From a cross-sectional analysis using national survey data from 1189 Jamaicans with creatinine measurements, an estimated 5% of the population had eGFR <60 mL/min/1.73 m.¹⁹⁸ In terms of pediatric CKD, congenital urologic kidney disease and glomerulonephritis are common causes of ESKF.¹⁹⁹

Kidney biopsy is available at one institution, and there is one nephro-pathologist, but there is no electron microscopy. In a biopsy series of 224 adults over a 3-year period, focal segmental glomerulosclerosis and lupus nephritis were the most common biopsy findings.²⁰⁰ Among 270 kidney biopsies performed in children spanning 25 years, diffuse glomerulonephritis and membranoproliferative glomerulonephritis were the most common findings. Most (53%) were idiopathic, with 32.3% infection related (mainly poststreptococcal glomerulonephritis [PSGN], followed by hepatitis B). However, there was a noticeable decline in PSGN rates in the study period. Educational initiatives from pediatric nephrologists targeting early recognition and management of PSGN by general pediatricians may have explained this decline.²⁰¹

KIDNEY REPLACEMENT THERAPY IN JAMAICA

The health care model of Jamaica is mixed public and private, but private care is outsourced where public services are overwhelmed. In the case of dialysis, due to limited accessibility of public units, most dialysis is sourced privately. In-center hemodialysis is the predominant form of renal replacement. Most patients are underdialyzed with two-thirds dialyzed twice weekly. Jamaica has one of the highest out-of-pocket costs for hemodialysis in the Caribbean, with each treatment costing approximately US\$90 to US\$117 per session and less than one-fifth of the Jamaican population having insurance.¹⁹⁷ PD is available for adults and children in two public institutions and one semiprivate institution. The first kidney transplant was performed in Jamaica in the 1980s, but kidney transplantation sustainability has been challenged by the cost and availability of immunosuppressive

medications. Efforts for kidney transplantation program have been described earlier.

SCOPE FOR RESEARCH AND INNOVATION

The Caribbean represents a unique diaspora, with social, environmental, and demographic exposures that influence kidney health. Despite a preponderance of kidney disease risk factors, several gaps in the knowledge of the rates, characteristics, and outcomes of both acute and CKD exist. Given the trajectory of noncommunicable disease and changes in the sociodemographic profile of Caribbean countries, epidemiologic data on kidney disease will be needed to guide policy and resource allocation to improve equity in kidney disease care in this region. With the advent of increasing use of precision medicine in renal disease, knowledge of kidney disease causative mechanisms, biomarkers, and therapeutic targets, the scope of kidney medicine worldwide is evolving. The Caribbean represents an uncharted opportunity for observational longitudinal studies exploring the effect of climate, socioeconomic, and genetic factors on incident CKD and its progression and clinical trials on biomarkers and therapeutics in kidney disease.

CONCLUSION

The burden of CKD is increasing across the small Island States of the Pacific and Caribbean, driven by inexorable growth in noncommunicable chronic illnesses, such as diabetes mellitus. Access to health care and data on health outcomes or burden of disease is highly variable and mostly limited in many nations of these two regions outside of Australia and NZ. Even within affluent countries, such as Australia and NZ, important inequities in health outcomes affect the indigenous populations. Factors that perpetuate health inequalities include the ongoing impact of colonization and racism, as well as socioeconomic factors and regional isolation. In response to growth in the prevalence of CKD and its associated adverse health outcomes, various governments are currently expanding their capacities to improve care for these patients. Implementing infrastructure to capture the true disease burden and health outcomes will be an important component of strategies to achieve better health outcomes.

Both the Caribbean and the Pacific represent a unique diaspora, with social, environmental, and demographic exposures that influence kidney health. Despite a preponderance of kidney disease risk factors, several gaps in the knowledge of the rates, characteristics, and outcomes of both acute and chronic kidney disease exist. Given the trajectory of noncommunicable diseases and changes in sociodemographic profiles, epidemiologic data on kidney disease will be needed to guide policy and resource allocation to improve equity in kidney disease care in these regions. With the advent of increasing use of precision medicine in renal disease and knowledge of kidney disease causative mechanisms, biomarkers, and therapeutic targets, the scope of kidney medicine worldwide is evolving. The Caribbean and Pacific represent an uncharted opportunity for observational longitudinal studies exploring the effect of climate, socioeconomic factors, the impact of

colonization and genetic factors on incident CKD, and its progression and clinical trials on biomarkers and therapeutics in kidney disease.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the substantial contributions of the entire Australia and NZ nephrology community in providing information for and maintaining the ANZDATA database. We also thank the ANZDATA executive for granting permission to use figures from the ANZDATA Registry annual report.

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QUESTIONS

- With respect to kidney disease in Oceania, which of the following is TRUE?
 - Chronic glomerulonephritis is the commonest cause of chronic kidney disease in Oceania.
 - Alport syndrome is the most common monogenic inherited disorder in French Polynesia.
 - IgA nephropathy is the commonest form of primary glomerular disease in New Caledonia.
 - The prevalence of diabetic kidney disease in the Pacific Islands is lower than that in Australia and New Zealand.
 - Leptospirosis is an uncommon cause of acute kidney injury in Fiji.

Answer: b

Rationale: Alport syndrome is the most prevalent monogenic inherited disorder found within French Polynesia, accounting for 18% of the patients receiving dialysis (compared with 1%–2% of end-stage kidney disease patients in Europe and 2.3% of renal transplant patients in the United States). Diabetic kidney disease is the commonest cause of chronic kidney disease in Oceania and is more prevalent in the Pacific Islands than in Australia and New Zealand. The prevalence of diabetic kidney disease in Fiji is one of the highest in the world. The most prevalent primary glomerular disease in New Caledonia is focal segmental glomerulosclerosis. Leptospirosis is a common cause of acute kidney injury in Fiji and most commonly affects young adult males, particularly following rainy periods.

- Which of the following statements about renal replacement therapy in Oceania is FALSE?
 - Australia has the highest rate of home hemodialysis in the world.
 - Kidney transplantation is generally not available in the Pacific Islands.
 - The prevalence of treated end-stage kidney disease in Australia and New Zealand is less than half that of the United States.
 - Approximately 50% of people who die from end-stage kidney disease in Australia do so without accessing renal replacement therapy.
 - Hemodialysis session durations in Australia and New Zealand are on average more than an hour longer than those in the United States.

Answer: a

Rationale: New Zealand has the highest rate of home

hemodialysis in the world. Australia has the second highest rate of home hemodialysis in the world. All other statements are true.

- Which of the following statements about ethnic inequalities in Australia is FALSE?
 - The proportion of indigenous Australians with end-stage kidney disease due to diabetic kidney disease is approximately double that of nonindigenous Australians.
 - The hospitalization rate for regular dialysis treatment in indigenous Australians is 11 times as high as for other Australians.
 - Hospitalization rates for acute kidney injury in indigenous Australians are almost three times higher than those of nonindigenous Australians.
 - Indigenous Australians are less likely to receive a kidney transplant than nonindigenous Australians.
 - Indigenous Australians living in urban locations have a higher risk of developing end-stage kidney disease than those living in rural and remote locations.

Answer: e

Rationale: Indigenous Australians living in rural and remote locations have a markedly increased risk of developing end-stage kidney disease compared with those living in urban locations. All other statements are true.

- Which of the following statements about renal replacement therapy for ESKF care in the Caribbean is true?
 - Kidney transplantation is not available in the Caribbean.
 - Peritoneal dialysis is the most common form of renal replacement therapy in the Caribbean.
 - In-center hemodialysis is the most common form of renal replacement therapy in the Caribbean.
 - Hemodialysis is available in all islands of the Caribbean.
 - Peritoneal dialysis is available in all islands of the Caribbean.

Answer: e

Rationale: Kidney transplantation is available in only a few countries (Puerto Rico, Trinidad, Tobago, Dominican Republic, and Cuba). Renal replacement therapy is available in most countries. In some countries, long-term renal replacement (either PD or HD) is not available and must be outsourced. In-center hemodialysis is the most common form of renal replacement in the region.